Rigidity of Cotangent Complex at Odd Primes

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1 Introduction

In his seminal work, Quillen proposed the following homology theory for augmented simplicial R-algebras, which is now dubbed Quillen homology: there is a Quillen adjunction

$$\mathrm{Ab} : \mathrm{sCAlg}_R^{\mathrm{aug}} \rightleftarrows \mathrm{sMod}_R : R \oplus (-)$$

For an object $X \in \mathrm{sCAlg}_R^{\mathrm{aug}}$, the Quillen homology of X is defined to be the module homology of $\mathbb{L}\mathrm{Ab}(X)$. With this construction, Quillen gave the definition of the cotangent complex $\mathbb{L}_{A/R}$ associated to morphism of commutative rings $R \to A$. let $P \to A$ be a cofibrant replacement of A in $s\mathbf{CAlg}_R$. The cotangent complex $\mathbb{L}_{A/R}$ is modelled by the simplicial A-module

$$\mathbb{L}_{A/R} := \Omega_{P/R} \otimes_P A$$

By the Dold-Kan correspondence, we will use $\mathbb{L}_{A/R}$ to denote both the simplicial A-module and its normalization in $\mathrm{Ch}^+(A)$.The Quillen homology is defined to be

$$H_i^Q(A/R; k) := \pi_i(\mathbb{L}_{A/R} \otimes_A k) \cong \mathrm{Tor}_i^A(\mathbb{L}_{A/R}; k)$$

In his seminal work, Quillen also proposed the following rigidity conjecture,

Conjecture 1.0.1. If $\mathrm{pd}(\mathbb{L}_{A/R}) < \infty$, then $\mathrm{pd}(\mathbb{L}_{A/R}) \leq 2$.

2 Outline

Fix a morphism of Noetherian local k-algebras $R \to A$. Standard localization techniques reduces the conjecture to the case where we consider a surjection of Noetherian local k-algebras $(R, \mathfrak{m}) \to (A, \mathfrak{n})$. In this case, the flat dimension of the cotangent complex can be detected by tensoring with the residue field

Proposition 2.0.1 (Stack 15.79.2)**.** Let R be a ring. Let $a, b \in \mathbb{Z}$. Let $K^\bullet$ be a pseudo-coherent complex of R-modules. The following are equivalent:

1. $K^\bullet$ is perfect with Tor amplitude in $[a, b]$.
2. For every maximal ideal $\mathfrak{m}$, we have $\mathrm{Tor}_i^R(K^\bullet; k(\mathfrak{m})) = 0$ for $i \notin [a, b]$

Thus, under perfectness assumption, it suffices to show $H_i^Q(A/R; k) = 0$ for $i > 2$. We will therefore suppress the coefficient k in Quillen homology from now on, unless other coefficients are used.

To prove the conjecture, Turner's program utilizes many tools from the homotopy theory of simplicial commutative algebras. For a k-vector space V, let $K(V, n)$ be the simplicial k-vector space characterized by having homotopy group concentrated in degree n, with value V.

Theorem 2.1. Given $A \in s\mathbf{CAlg}_k$, which is connected, there exists a sequence of cofibrations

$$A = A(0) \xrightarrow{j_0} A(1) \xrightarrow{j_1} A(2) \xrightarrow{j_2} ...$$

with the following properties

1. For each $n \geq 1$, $A(n)$ is $(n-1)$-connected.
2. For $s \geq n$,
$$H_s^Q(A) \cong H_s^Q(A(n))$$
3. There is a cofiber sequence
$$\mathrm{Sym}_k(K(H_n^Q(A), n)) \to A(n) \to A(n+1)$$

In particular, for A such that $H_i^Q(A)$ vanishes for $i > n$, we may identify

$$A(n) \cong \mathrm{Sym}_k(K(H_n^Q(A), n))$$

and obtain a comparison map

$$\phi_n : A \to \mathrm{Sym}_k(K(H_n^Q(A), n))$$

inducing isomorphism on H_n^Q. The work of Bousfield, Goerss and Lada identifies the homotopy groups of the right hand side, which is generated by compositions of homotopy operations.

Theorem 2.2 (Bousfield, 8.11)**.** Let $\{a\}_{a\in A}$ be $\mathbb{F}_p$-basis for $\pi_* X$. Then $\oplus_{s=0}^{\infty}\pi_t(\mathrm{Sym}^s X)$ is the non-unital associative graded-commutative $\mathbb{F}_p$-algebra with generators

$$\{\alpha_I(a) | a \in A, I \text{ admissible, } \mathrm{excess}(I) \leq |a|\}$$

In particular, one could find a composition of homotopy operations, of arbitrary length, acting non-trivially on the degree n-element in $\pi_*(\mathrm{Sym}_k(K(H_n^Q(A), n)))$. Now the proof reduces to two parts:

Theorem 2.3. Suppose $n > 2$. There is non-zero element in $y \in \pi_* A$ such that $(\phi_n)_*(y) = a$, and one could find a composition of homotopy operations, of arbitrary length, acting non-trivially on y.

The proof of the theorem is comparing the Quillen spectral sequences using ϕ_n.

Theorem 2.4. There is a convergent spectral sequence

$$E^1_{s,t} := \pi_t(\mathrm{Sym}^s(H_*^Q(A/R; k))) \Rightarrow \pi_{s+t} X$$

This is the step where Turner had to restrict to characteristic 2: he needed explicit Adem relations, which is still absent to this day, among operations to control the differentials.

On the other hand, for $A = P \otimes_R k$, operations on $\pi_* A$ has the following nilpotence property

Theorem 2.5. There exists some integer N such that any composition of strictly suspended operation of length $> N$ on $\pi_* A$ is zero.

This is where Turner needed the assumption that A is Cohen-Macauley, so he could reduce to the Artinian case where the maximal ideal is nilpotent.

3 Homotopy Operations on Simplicia $\mathbb{F}_p$-Algebras

Theorem 3.1 (Bousfield, 8.11)**.** For an odd prime p, and a simplicial $\mathbb{F}_p$-algebra B, there exists two families of operations

$$v_t : \pi_n B \to \pi_{n+2t(p-1)} B \qquad 2 \leq 2t \leq n$$
$$\eta_t : \pi_n B \to \pi_{n+2t(p-1)+1} B \qquad 3 \leq 2t+1 \leq n$$

The operations have the following properties

1. The operations satisfy
$$v_t(a \cdot u) = a^p u$$
for $a \in \pi_0 B$
2. Except the top operation, all operations are additive. The top operation is the divided power.

Definition 3.1.1. For $k \geq 0$, let $I = (\delta_1, s_1, ..., \delta_k, s_k)$ be a $(2k)$-tuple of integers, where $\delta_i = 0, 1$ and $s_i \geq 1$. The index I is called **admissible** if $s_i \geq ps_{i+1} + \delta_i$ for $1 \leq i \leq k-1$. Its **excess** is defined to be

$$\text{excess}(I) := \begin{cases} 0 & k = 0 \\ 2s_1 + \delta_1 & k = 1 \\ 2s_1 + \delta_1 - \sum_{i=2}^{k}(2(p-1)s_i + \delta_i) & k \geq 2 \end{cases}$$

Define

$$\alpha_{\delta_i, s_i} := \begin{cases} v_{s_i} & \delta_i = 0 \\ \eta_{s_i} & \delta_i = 1 \end{cases}$$

and

$$\alpha_I := \alpha_{\delta_1, s_i} \circ ... \circ \alpha_{\delta_k, s_k}$$

If out of range just zero

Theorem 3.2 (Bousfield, 8.11)**.** Let $\{a\}_{a \in A}$ be $\mathbb{F}_p$-basis for $\pi_* X$. Then $\oplus_{s=0}^{\infty} \pi_t(\text{Sym}^s X)$ is the non-unital associative graded-commutative $\mathbb{F}_p$-algebra with generators

$$\{\alpha_I(a) | a \in A, I \text{ admissible}, \ \text{excess}(I) \leq |a|\}$$

In particular, the generators provide an infinite tower of non-zero operations.

Corollary 3.2.1. Let V be a k-vector space, and let u be the generating class in $\pi_n(\text{Sym}_k K(V, n))$. For any $j > 1$, the index $I_j := (0, p^j, 0, p^{j-1}, ..., 0, p, 0, 1)$ is admissible, and thus the operation $\theta_j := \alpha_{I_j}$ is non-zero on u for all j.

Lemma 3.3 (Degree-Weight Bound)**.** Let X be a simplicial k-module.

1. Suppose $d = \dim \pi_1(X) < \infty$. Then $\pi_t(\text{Sym}^s X)$ is zero if $t < 2s - d$.
2. Suppose $X = K(V, n)$. Then $\pi_t(\text{Sym}^s X)$ is zero if $t < 2s + n - 2$.

Proof. Let y be a non-zero monomial in $\pi_*(\text{Sym}_k X)$. By Theorem 3.2, we can express y as product of k admissible monomial generators

$$y = \alpha_{I_1}(x_1)...\alpha_{I_k}(x_k)$$

where each I_j is admissible. Let $I_j := (\delta_1, s_1, ..., \delta_n, s_n)$. Admissiblity provides a lower bound on degree in terms of the weight

$$\begin{aligned} \deg(\alpha_{I_i}(x_i)) &= \deg(x_i) + 2(p-1)\sum_{i=0}^{n-1} s_i \\ &\geq \deg(x_i) + 2(p-1)\sum_{i=1}^{n-1} p^i \\ &= \deg(x_i) + 2(p^n - 1) \\ &= \deg(x_i) + 2\text{wt}(\alpha_I) - 2 \end{aligned}$$

Since degree and weight both add in the product, we get

$$\deg(y) \geq 2\text{wt}(y) + \sum_{i=1}^{k} \deg(x_i) - 2$$

If $X = K(V, n)$, then $|x_i| = n$ for all i, and part (2) follows easily.

If $d = \dim(H_1^Q(Q)) < \infty$. We see only $|x_i| = 1$ will contribute to a negative summand, and each basis vector can only appear once in the product for it to be non-zero by graded commutativity. Therefore

$$\deg(y) > 2\mathrm{wt}(y) - d$$

so part (1) follows.

□

4 Detection in Quillen Spectral Sequence

Let A be an augmented simplicial $\mathbb{F}_p$-algebra. The Quillen spectral sequence reads

$$E^1_{s,t} := \pi_t(\mathrm{Sym}^s(H_*^Q(A))) \Rightarrow \pi_{s+t}X$$

Suppose n is the top non-vanishing degree of $H^Q(A)$, and let $u \in H_n^Q(A)$ be the generating class. Then, the element θ_j defined in Corollary 3.2.1 is an element in bidegree $(s,t) := (p^j, n + 2(p^j - 1))$. We can analyze the permanence property of this cycle:

Proposition 4.0.1. The element θ_j can be represented by a cycle $a \in F^sN(P)_t$, and $da \in F^{2s}N(P)_{t-1}$.

In particular, the spectral sequence differential d_r is trivial on the representing class a for $r < s$, as the target $F^{2s}N(P)_{t-1} \subseteq F^{s+r+1}N(P)_{t-1}$ is trivial in the subquotient $\frac{F^{s+r}N(P)_{t-1}}{F^{s+r+1}N(P)_{t-1}}$.

Proposition 4.0.2. The Quillen spectral sequence of $\mathrm{Sym}K(H_n^Q(A), n)$ degenerates on E^1.

For $r \geq s$, the target groups will be a subquotient of

$$E^1_{s+r,2s+n-3}$$

By Lemma 3.3, we see the target groups automatically vanish once we choose j large enough.

On the other hand, the there are no incoming differentials before page s, by comparing the spectral sequence with that of $\mathrm{Sym}K(H_n^Q(A), n)$. There are no incoming differentials on or after page s simply by degree reasons.

extension: by comparing the spectral sequences, we see that there is no extension problems: the difference lands in $\pi_{n+2s-2}\mathrm{Sym}^{s+1}(K(H_n))$. This group is zero by part (2) of Lemma 3.3.

5 Nilpotence of Operations on Local Derived Fiber

Let $(R, \mathfrak{m})$ be Noetherian local ring, M a finitely generated R-module. Choose a resolution of M be finite free R-modules

$$\ldots \xrightarrow{d_2} F_1 \xrightarrow{d_1} F_0 \xrightarrow{d_0} M \to 0$$

Theorem 5.1 (Uniform Artin–Rees for Syzygies)**.** For a fixed ideal $J \subseteq R$, there is a uniform integer $h \geq 0$ such that for every $n \geq h$, and $i \geq 0$, one has

$$J^n F_i \cap \mathrm{im}(d_{i+1}) \subseteq J^{n-h}\mathrm{im}(d_{i+1})$$

Take $J = \mathfrak{m}$, one has the following corollary

Corollary 5.1.1. There is a uniform integer $h \geq 0$, such that for every $q \geq h$ and $i \geq 0$, the reduction

$$\mathrm{Tor}_i^R(M; R/\mathfrak{m}^q) \to \mathrm{Tor}_i^R(M; k)$$

is zero.

Theorem 5.2 (Nilpotence on Local Derived Fiber)**.** There is an integer l such that all composition of operations of length greater than l vanishes on $\pi_*(P \otimes_R k)$.

Proof. An element $a \in N(P) \otimes_R k$ is a cycle iff $da \in \mathfrak{m}N(P)$. Let α_I be a composition of a length k operation of the form v_t. Once we show compatible operations on chain also satisfy

$$d\alpha_I(a) \sim \alpha_I(d(a)) \in \mathfrak{m}^{p^t} N(P)$$

It follows that a is the image of the reduction map, and by the previous corollary, we know there is a uniform integer such that the reduction map is zero. □

6 Chain Model

Since the spectral sequence is defined via filtration on chains, we need to have a chain representative of the operations, which a priori is defined via universal homotopy classes, in order to analyze how the operations interact with the differentials. We begin with a quick review of the operations.

Suppose we are looking for homotopy operations on $\pi_* B$ for some simplicial k-algebra B

$$\pi_n B \to \pi_{n+t} B$$

By representability and adjunction, these operations are represented by classes in $\pi_{n+t}(\mathrm{Sym}K(k,n))$. Let $a : K(k,n) \to B$ be a representative of some class $\tilde{a} \in \pi_n B$, and $u : K(k, n+t) \to \mathrm{Sym}(K(k,n))$ to be a representative of $\tilde{u} \in \pi_n B$. The evaluation $u(a) \in \pi_{n+t} B$ is given by the composition

$$K(k, n+t) \xrightarrow{u} \mathrm{Sym}(K(k,n)) \xrightarrow{a} B$$

Under the Dold-Kan correspondence, operations can be understood more explicitly: they are classes in $H_{n+t}(N(\mathrm{Sym}K(k,n)))$, which can be represented by some chain map

$$k[n+t] \to N(\mathrm{Sym}K(k,n))$$

which corresponds to an explicit degree $n+t$ cycle $z_{n+t} \in N(\mathrm{Sym}K(k,n))$.

6.1 Suspension on Chain Complexes

Let C be the chain complex concentrated in degree i. There is an exact sequence of chain complexes

$$0 \to C \to D \to C[1] \to 0$$

where $D := \mathrm{Cone}(\mathrm{id}_C)$ is the cone, which is acyclic. Then the connecting morphism

$$\partial : H_j(C[1]) \to H_{j-1}(C)$$

is the suspension isomorphism. In the case $C = k[n-1]$ and under Dold-Kan, the exact sequence becomes

$$0 \to K(k, n-1) \to K(D_n) \to K(k,n) \to 0$$

We shall refer to these as **additive suspension**. The non-additive case happens when we apply the reduced symmetric algebra functor: we have the commutative diagram of simplicial commutative non-unital algebras

$$\begin{array}{ccccc} \mathrm{Sym}K(k,n-1) & \xrightarrow{i_*} & \mathrm{Sym}K(D_n) & \longrightarrow & \mathrm{Coker} \\ & & \downarrow q & \nearrow & \\ & & \mathrm{Sym}K(k,n) & & \end{array}$$

Note that $\mathrm{Sym}K(D_n)$ is still acyclic. After normalization, a class $z \in N(\mathrm{Sym}K(k, n-1))$ still injects to a class $i_*(z)$, which is a boundary. Choose ω such that $d\omega = i_*(z)$, and define

$$\sigma z = q(w) \in N(\mathrm{Sym}K(k, n+1))$$

It is clear that $q(w)$ is a cycle since w is coming from the kernel, and the choice of ω does not affect the homology class of σz.

Theorem 6.1 (Bousfield)**.** Suppose z_{pn} is the normalized cycle representing the top divided power operation, coming from the universal generator in

$$v_n \in \pi_{pn}(\mathrm{Sym}^p K(k, n))$$

Then, σz_{pn} represents the additive operation from the generator

$$\pi_{pn+1}(\mathrm{Sym}^p K(k, n+1))$$

The input degree and output degree are both raised by 1, and the resulting operations is no longer the top operation. In other words, all the additive operations in v_t family are all suspended from the top divided powers from lower input degree.

Lemma 6.2 (Chain Lift with Controlled Differential)**.** Let B be a simplicial commutative $\mathbb{F}_p$ algebra. Let $a \in N(B)_n$ and $2j < n$. There exists a chain lift

$$V_j^n(a) \in N(P)_{n+2j(p-1)}$$

satisfying

1. If a is a cycle, then $V_j^n(a)$ represents $v_j(a)$.
2. If P is an algebra over discrete commutative ring R, $J \subseteq R$ an ideal, and $da \in J^m(N(B))_{n-1}$

$$dV_j^n(a) \in J^{pm}(N(B))_{(n-1)+2j(p-1)}$$

3. If P is augmented over k, with augmentation ideal I, and $a \in N(I^s)_n$ with $da \in N(I^t)_{n-1}$ for some $t \geq s$, then
$$V_j^n(a) \in N(I^{ps})_{n+2j(p-1)}, \qquad dV_j^n(a) \in N(I^{pt})_{(n-1)+2j(p-1)}$$

Proof. Let v be the generator in n, and u be the generator in degree $n-1$ of D_n, with $dv = u$. There is a simplicial k-module map

$$D_n \to N(B)$$

by $v \mapsto a$, $u \mapsto da$. Since $da = 0$, we see the induced normalized algebra map a_* factors through q

$$\begin{array}{ccc} N(\mathrm{Sym}K(k, n-1)) & & \\ \downarrow{\scriptstyle i_*} & & \\ N(\mathrm{Sym}K(D_n)) & \xrightarrow{a_*} & N(B) \\ \downarrow{\scriptstyle q} & \overset{\tilde{a}}{\nearrow} & \\ N(\mathrm{Sym}K(k, n)) & & \end{array}$$

Let $z_j^n \in N(\mathrm{Sym}K(k, n))_{n+2j(p-1)}$ be the cycle representing the operation v_j. By Theorem 6.1, z_j^n is suspended from some $z_j^{n-1} \in N(\mathrm{Sym}K(k, n-1))_{(n-1)+2j(p-1)}$. Choose ω_j^n such that $d(\omega_j^n) = z_j^{n-1}$, with $q(\omega_j^n) = z_j^n$, we define

$$V_j^n(a) := a_*(\omega_j^n)$$

It is immediate from construction that $V_j^n(a)$ represents the class $v_j(a)$. □

A Quillen Spectral sequence

By the standard model structure on $s\mathbf{CAlg}_R$, one may choose a cofibrant replacement $P \to A$ such that P is semi-free, i.e every P_n is a free R-algebra. Let $X := P \otimes_R k$. Denote by I the augmentation ideal and $QX := I/I^2$ the indecomposables. Standard computation gives

$$QX \cong \Omega_{P/R} \otimes_P k$$

And therefore we have $H_i^Q(A/R) \cong \pi_i(QX)$. This provides the interpretation that Quillen homology of an augmented simplicial k-algebra X is the homotopy group of its indecomposables QX. we have a descending filtration of X by

$$X \supseteq I \supseteq I^2 \supseteq I^3 \supseteq ...$$

which leads to a spectral sequence

$$E^1_{s,t} := \pi_t(I^s/I^{s+1}) \Rightarrow \pi_{s+t}X$$

A standard fact on the model structure on $s\mathbf{CAlg}_R$ says one may choose P to be semi-free, meaning each simplicial degree P_n is a free R-algebra. It follows that

$$I^s/I^{s+1} \cong \mathrm{Sym}^s_k(I/I^2)$$

Since a chain complex of vector spaces splits as direct sums of its homology, I/I^2 also splits as the direct sum of its homotopy groups under the Dold-Kan correspondence.